AGILE DEVELOPMENT PROCESS AND DEVOPS LAB

FAT LAB

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Online Auction System

Problem Statement:

Ensure bids are recorded and highest bidder is selected correctly.

Tasks:

1. Develop bidding logic

```
PS D:\online-auction-system> docker run --rm online-auction-system:latest
=== Online Auction System ===

Auction created: Rare Painting (ID: ART001)

--- Bidding Starts ---
✓ Bid placed: Alice -> $150.0 on Rare Painting
✓ Bid placed: Bob -> $200.0 on Rare Painting
✗ Bid rejected: $175.0 is below minimum bid
✓ Bid placed: Alice -> $550.0 on Rare Painting

📊Current highest bid: Bid{bidder='Alice', amount=$550.00, time=2026-04-15T12:57:04.326343499}

--- Auction Closing ---
🏆Auction closed! Winner: Alice with $550.0

🎉WINNER: Alice with $550.0

PS D:\online-auction-system>
```

2. Write JUnit tests for bid validation

```
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.auction.AuctionServiceTest
Auction created: Test Item (ID: TEST001)
? Bid placed: Alice -> $100.0 on Test Item
? Bid placed: Bob -> $250.0 on Test Item
? Bid placed: Charlie -> $300.0 on Test Item
Auction created: Test Item (ID: TEST001)
? Bid placed: Alice -> $400.0 on Test Item
Auction created: Test Item (ID: TEST001)
? Bid rejected: $50.0 is below minimum bid
Auction created: Test Item (ID: TEST001)
Auction created: Test Item (ID: TEST001)
Auction created: Test Item (ID: TEST001)
? Bid placed: John -> $150.0 on Test Item
Auction created: Test Item (ID: TEST001)
? Bid placed: Alice -> $500.0 on Test Item
[INFO] Tests run: 7, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.163 s - in com.auction.AuctionServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 7, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  4.855 s
[INFO] Finished at: 2026-04-15T18:27:48+05:30
[INFO] ------------------------------------------------------------------------

PS D:\online-auction-system>
```

3. Implement CI/CD pipeline with automated testing

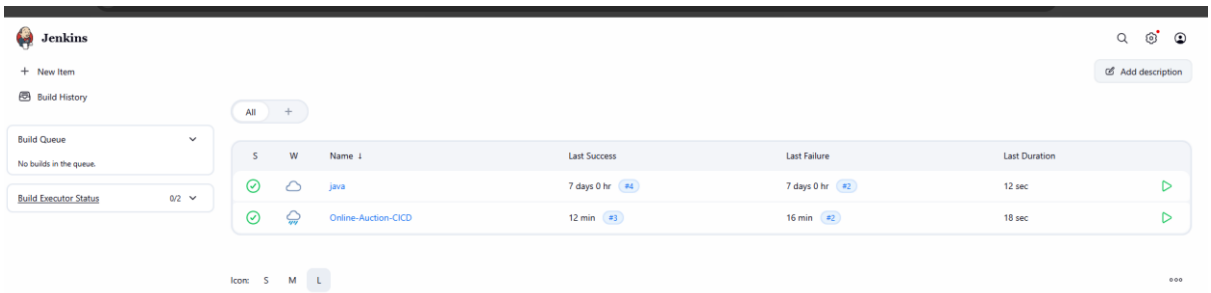
➢ Set up version control using GitHub

➢ Configure CI pipeline (e.g., Jenkins/GitHub Actions) to trigger on code commits

➢ Automate build process using Maven

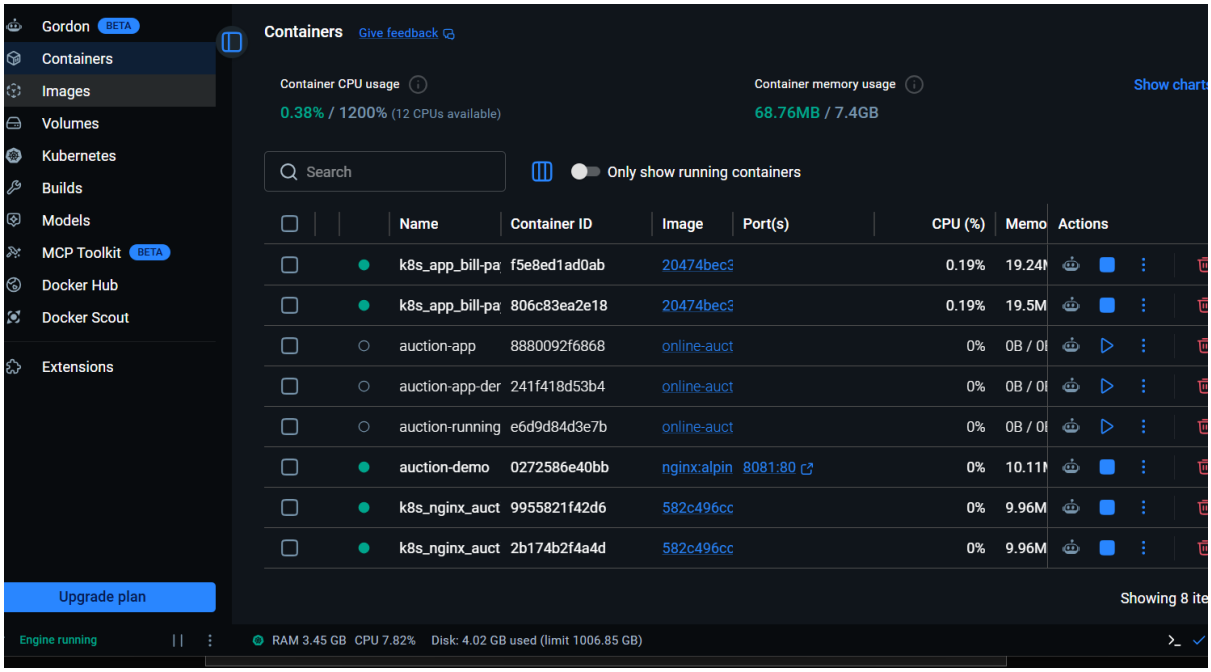
➢ Integrate automated unit tests in pipeline

➢ Configure CD pipeline for automatic deployment to staging/production

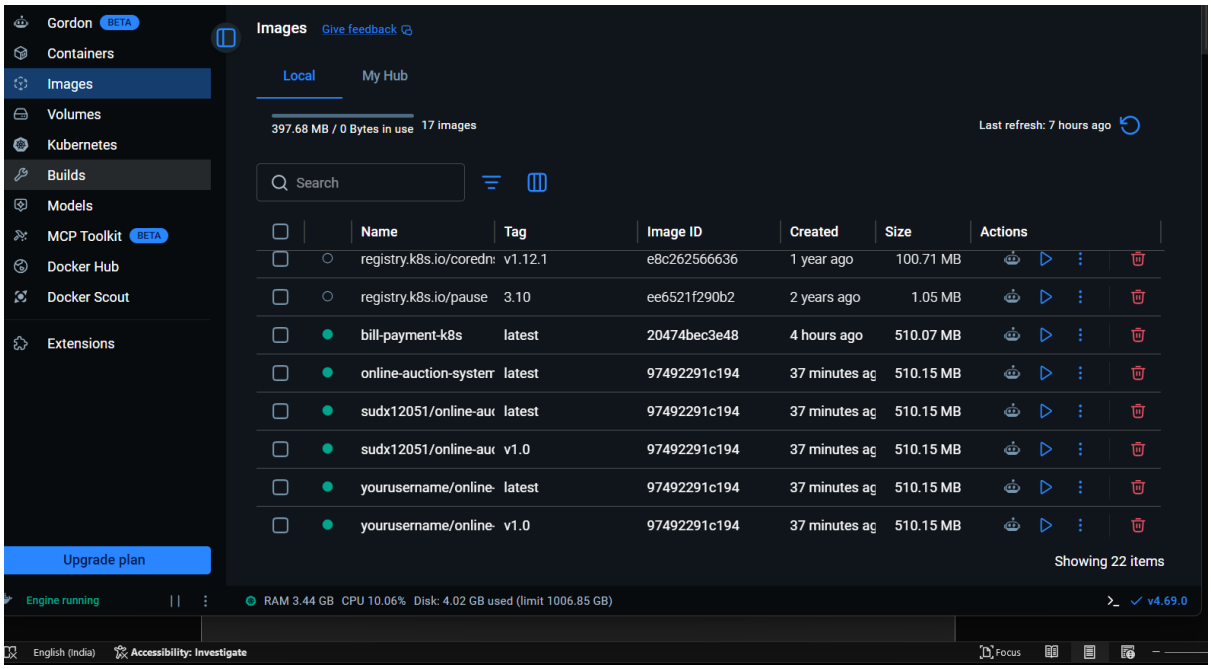


4. Deploy services on Docker and Kubernetes

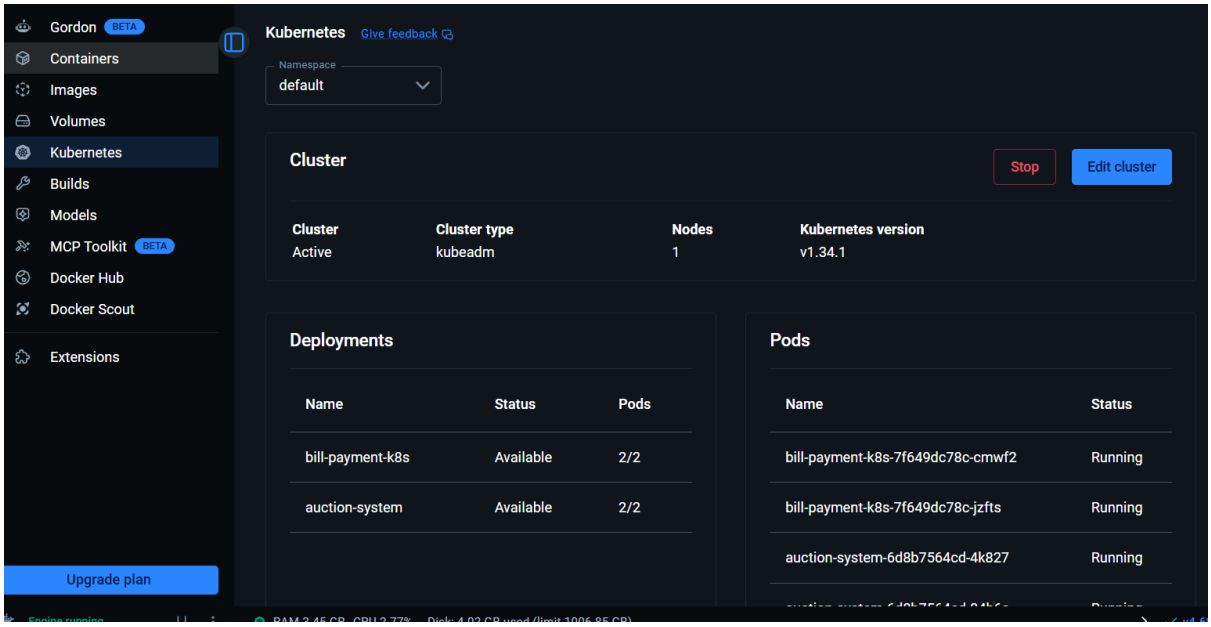
➢ Containerize application using Docker (write Dockerfile)



➢ Build and push Docker image to container registry (Docker Hub)



➢ Create Kubernetes deployment and service YAML files


Gordon BETA
Containers
Images
Volumes
Kubernetes
Builds
Models
MCP Toolkit BETA
Docker Hub
Docker Scout
Extensions
Upgrade plan
Kubernetes
Give feedback
Namespace
default
Cluster
Stop
Edit cluster
Cluster
Active
Cluster type
kubeadm
Nodes
1
Kubernetes version
v1.34.1
Deployments
Name
Status
Pods
bill-payment-k8s
Available
2/2
auction-system
Available
2/2
Pods
Name
Status
bill-payment-k8s-7f649dc78c-cmwf2
Running
bill-payment-k8s-7f649dc78c-jzfts
Running
auction-system-6d8b7564cd-4k827
Running